TurboQuant v1.5.3

Work Log — Side Buffer Experiment + Double WHT Integration

2026-04-11 ~ 04-12 | GPT-OSS 120B · head_dim=64

f16 Side Buffer 35/35 | src1 인덱스 버그 발견 | 3+1비트 Idea

TBQ4 35/35 검증 | AmesianX

Session Overview

GOAL

Improve TBQP3_3 (3-bit K) math precision.
Achieve f16 K equivalent quality.

RESULTS

Side buffer experiment achieved 35/35.
TBQP3_3 original: Korean ✅, math ❌
TBQ4 recommended: 35/35 ✅

KEY DISCOVERY

src1 index mapping — side buffer must use same KV position mapping as set-rows.

IDEA

3+1 bit split — VRAM 3-bit + system RAM 1-bit = 4-bit quality.

f16 Side Buffer Experiment 35/35

> Intercept original K as f16 before quantization into a separate buffer. Attention reads f16 instead of quantized data.

- **Phase 1~3:** WHT 도메인 f16 → raw f16 → generation 포함 → 수제 dot product
- **Phase 4~5:** sharpening, *D/D 보상, Q WHT 스킵 → 효과 미미
- **Phase 6:** f16 커널 redirect (fattn.tu에서 K→F16 override) → 거의 성공
- **Phase 7:** 값 비교 디버깅 → **src1 인덱스 누락** 발견!
- **Phase 8:** src1 적용 → **35/35 math bench, Multi-turn 성공**

Root Cause: src1 Index BUG

set-rows stores at specific KV cache positions via `src1[i]`.
side buffer stored sequentially (0,1,2...) → **position mismatch**.

COMPONENT	BEFORE	AFTER
KV 캐시	src1[0]=5 → position 5	Same
Side buffer	**token 0 → position 0 ✗**	**token 0 → position 5 ✓**

Lesson: Measure, don't guess. Found by directly comparing f16 vs side buffer values.

3+1비트 분할 Idea 미구현

> VRAM 3-bit (engineering pride) + system RAM 1-bit = 4-bit quality

MODE	VRAM	RAM	QUALITY	SPEED IMPACT
3+0	3비트	0	cos 0.945	None
3+1	3비트	1비트	cos ~0.990	0.025ms
3+5	3비트	5비트	q8_0급	0.125ms
3+13	3비트	13비트	f16급	0.31ms

TBQ4 = 35/35 검증 → 3+1비트로 Same Quality 기대.
cudaHostAlloc zero-copy 사용.

Math Bench Results 35문제

CONFIG	K 캐시	V 캐시	MATH	한글	MULTI-TURN
f16/f16	f16	f16	**35/35**	✅	✅
side buffer	f16 (side)	tbq3_2	**35/35**	✅	✅
tbq4/tbq3	tbq4_2	tbq3_2	**35/35**	✅	✅
tbqp3/tbq3	tbqp3_3	tbq3_2	❌	✅	✅ **(9턴+)**

Recommended: `–cache-type-k tbq4 –cache-type-v tbq3` → 35/35 Math + 한글 + Multi-turn

v1.5.3 Code Changes

- **common.cpp:** head_dim=64 K → TBQP3_3 매핑 (QJL 재활성화)
- **fattn-vec.cuh:** is_cross_head_K=false, is_double_wht_K 플래그, Q Round 2, /D 보상
- **cpy-utils.cuh:** K 인코딩 double WHT per-head (512-point cross-head 폐기)
- **fattn.cu:** TBQ_TUNING D=64 전체 인스턴스 디스패치
- **CMakeLists.txt:** TBQ_TUNING D=64 인스턴스 파일 등록
- **ggml.h + 인스턴스 24개:** 주석 cross-head → double WHT per-head
- **논문 tex/PDF:** Cross-head 섹션 폐기, double WHT 교체

Git Status

브랜치	CONTENT	STATUS
main	v1.5.3 + upstream merge	**pushed**
experiment/side-buffer-f16	f16 side buffer 35/35	**pushed**
v1.5.3 태그	release tag	**pushed**

CI: CUDA 온리 워크플로우 (Windows CUDA 12.4/13.1 + Linux Docker CUDA 12.8)
아키텍처: sm_70 (V100), sm_80 (A100), sm_86 (3090Ti), sm_89 (4090), sm_90 (H100), sm_100/120 (5090)

Next Steps

TBQP3_3 MATH 개선

- Achieve matrix precision with 3-bit K
- Retune sharpening/softening
- Leverage side buffer experiment findings

3+1 BIT IMPLEMENTATION

- VRAM 3비트 + RAM 1비트
- cudaHostAlloc zero-copy
- Modify vec_dot (dual memory read)

MULTI-TURN 안정성

- Investigate channel tag parser issue
- LCP caching + side buffer integration

CI/RELEASE

- Verify workflow builds

- Binary release distribution

Speaker notes